



DMVPN



• GRE over IPsec의 한계점

- GRE는 p2p이므로 터널 개수가 많이 필요하다. 본사와 지사간 터널을 각각 만들고
 - 네트워크를 따로 할당해야 함
- 본사를 통해서만 통신이 가능

• DMVPN의 등장

- Point-to-Point 네트워크를 NBMA로 (본사-여러 지사 간 네트워크는 같은 대역)

• DMVPN의 요소

- mGRE(multipoint GRE)
- NHRP(next hop resolution protocol)
- IPSec
- Dynamic Routing

• DMVPN 버전

- PHASE 1 (nhrp v.1) : 본사는 mGRE, 지사는 P2P GRE
- PHASE 2 (nhrp v.2) : 본사와 지사 모두 mGRE, 지사간 통신 가능
- PHASE 3 : Hierarchy 구조, 사설 ip 사용 가능

• DMVPN

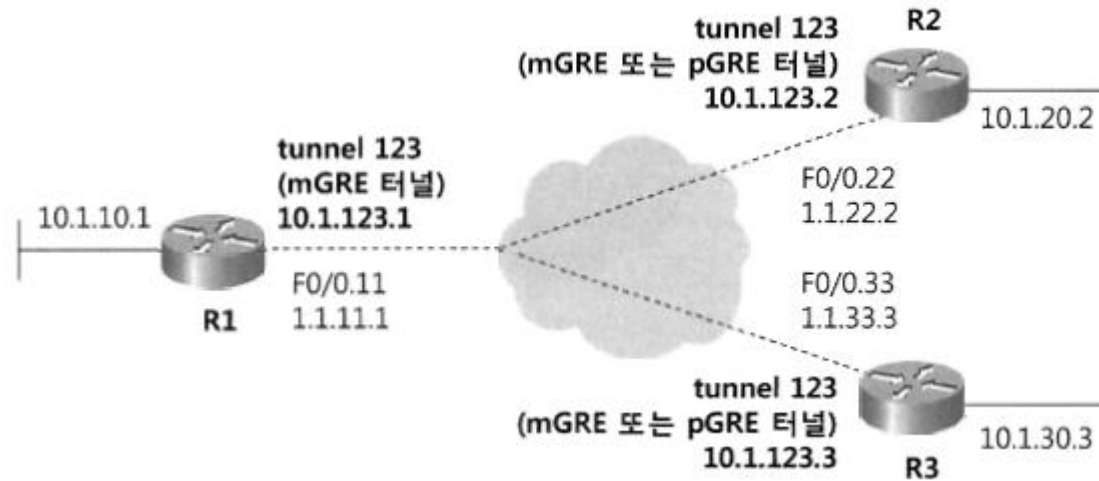
- -mGRE 터널과 NHRP를 사용하여 확장성 문제 해결
- 본사와 지사, hub & spoke 구조
- 본사를 통해서 지사간 통신이 가능
- 본사 장비로 목적지 네트워크와 연결되는 주소를 질의, 본사 장비가 목적지 네트워크와 연결되는 ip를 알려주고 이 후 두 개 지사간 직접 통신
- 데이터와 라우팅 정보 전송을 위해 mGRE 터널 사용

• DMVPN 장점

- 지사 장비는 다른 지사 장비와 GRE 터널 및 암호화 맵 설정할 필요가 없음
- 본사 장비는 지사 장비에 대한 상세 정보를 유지할 필요가 없다
 - 지사 장비 추가 시 본사 장비에 추가 설정 필요 없음
- 지사 장비는 자신의 외부 인터페이스 주소를 동적으로 할당 받을 수 있다
- 동적 라우팅 프로토콜 사용 → 복잡한 암호화 ACL이나 정적 라우팅 구성 필요 없음

• mGRE (multipoint GRE)

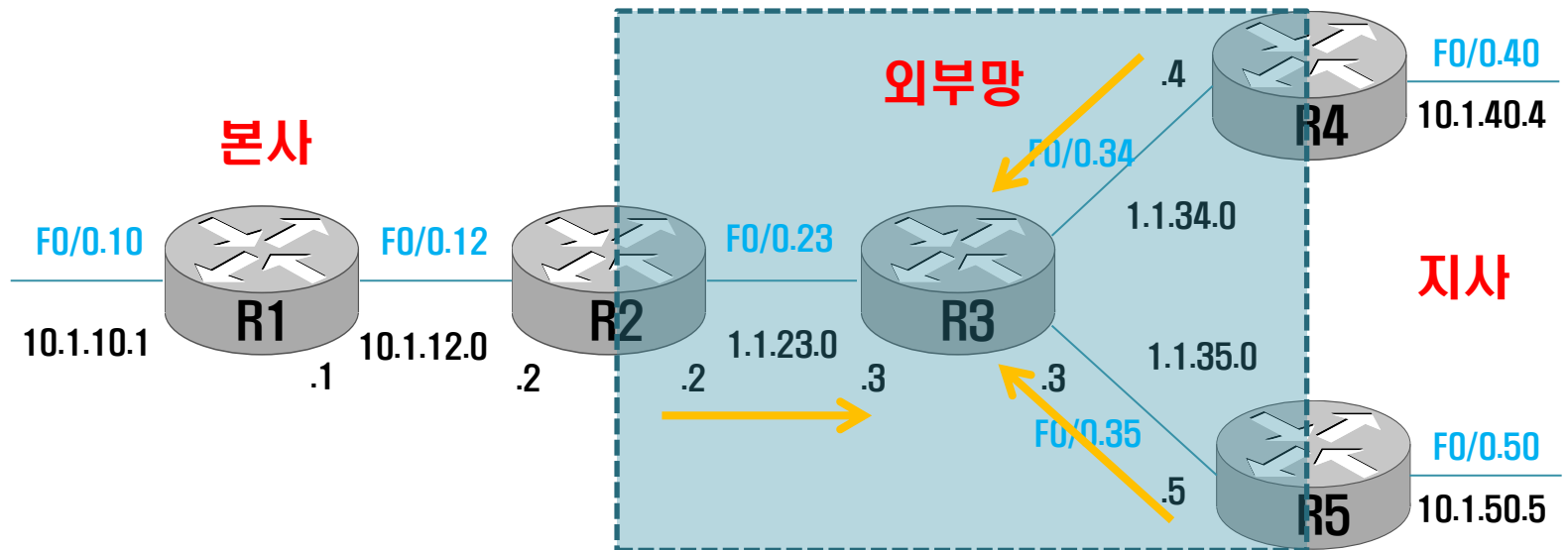
- p2p (point-to-point) GRE : 직접 연결되는 상대방이 하나인 GRE 인터페이스
- mGRE : 하나의 인터페이스를 통하여 직접 연결되는 상대방이 다수 개 있는 것
- 20byte 추가적인 ip 헤더, 4byte GRE 헤더, 4byte mGRE 터널키 사용



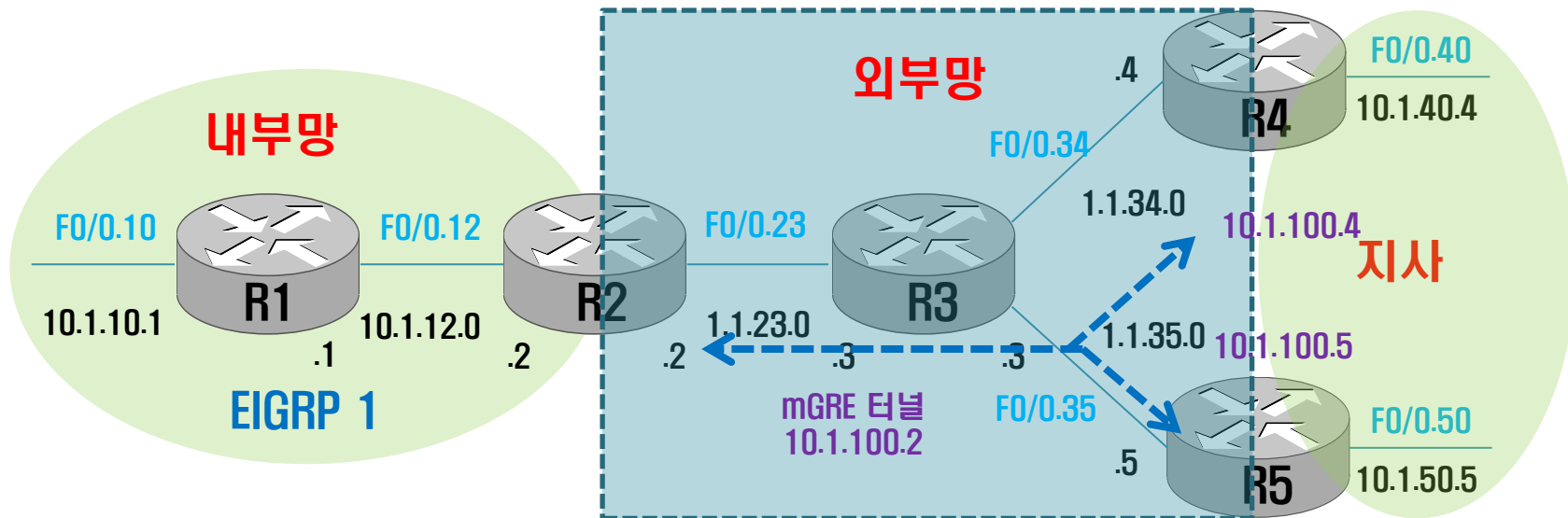
• NHRP

- 해당 목적지 ip로 가는 넥스트 홉 ip 주소를 알아내는 데 사용
- 동작절차
 - 1) 지사 라우터가 본사 라우터의 ip 주소로 자신의 ip 주소 등록
 - 2) 본사 라우터는 mGRE 인터페이스를 사용, 지사 라우터와 동일한 터널 구성
 - 3) NHRP는 터널 주소와 넥스트 홉 ip 주소를 매핑
- NHRP 등록과정에 참여하려면 모든 라우터가 네트워크 ID로 구분되는 동일한 NHRP 네트워크에 소속되어야 함, NHRP 네트워크 ID는 NHRP 도메인을 나타냄
- mGRE 터널에서는 다수개의 IPsec 피어가 존재하므로 넥스트홉 주소를 IPsec 피어 주소로 사용

- 테스트 네트워크 구축



- mGRE 터널 구성



• mGRE 터널 구성

//R2 - 본사설정

```
interface tunnel 0
 tunnel source 1.1.23.2
//터널 모드를 gre multipoint로 지정
 tunnel mode gre multipoint
 ip address 10.1.100.2 255.255.255.0
//동일 라우터에 다수개의 mGRE 인터페이스가 있을 때 구분하기 위한 터널 키 지정
 tunnel key 1
!
```

//NHRP 네트워크 ID지정, NHRP 도메인을 나타냄

```
ip nhrp network-id 1
```

//지사 라우터가 등록할 때 사용한 넥스트 홉 IP 주소로 멀티캐스트/브로드캐스트 패킷도 전송, 없으면 동적 라우팅 프로토콜 동작안함

```
ip nhrp map multicast dynamic
//NHRP NBMA 주소의 유효기간 표시
ip nhrp holdtime 600
!
```

//R4-지사 설정

```
interface tunnel 0
 tunnel source 1.1.34.4
 tunnel mode gre multipoint
 tunnel key 1
 ip address 10.1.100.4 255.255.255.0
!
```

//NHRP 서버 주소 지정

```
ip nhrp nhs 10.1.100.2
```

//목적지가 멀티캐스트,브로드캐스트 주소를 사용하는 패킷을 1.1.23.2로 전송

```
ip nhrp map multicast 1.1.23.2
```

//정적인 NHRP 매핑을 이용해 NHS의 넥스트 홉 IP 주소를 미리 지정, 터널 목적지 주소가 10.1.100.2인 패킷을 넥스트 홉 IP 주소 1.1.23.2로 전송

```
ip nhrp map 10.1.100.2 1.1.23.2
```

//NHS(next hop server)에게 60초마다 자신의 실제주소(NBMA 주소) 등록

```
ip nhrp registration timeout 60
!
```

• mGRE 터널 구성

//R5-지사 설정

```
interface tunnel 0
 tunnel source 1.1.35.5
 tunnel mode gre multipoint
 tunnel key 1
 ip address 10.1.100.5 255.255.255.0
 !
```

```
ip nhrp network-id 1
```

//NHRP 서버 주소 지정

```
ip nhrp nhs 10.1.100.2
```

//목적지가 멀티캐스트,브로드캐스트 주소를 사용하는 패킷을 1.1.23.2로 전송

```
ip nhrp map multicast 1.1.23.2
```

//정적인 NHRP 매핑을 이용해 NHS의 넥스트 홉 IP 주소를 미리 지정, 터널 목적지 주소가 10.1.100.2인 패킷을 넥스트 홉 IP 주소 1.1.23.2로 전송

```
ip nhrp map 10.1.100.2 1.1.23.2
```

//NHS(next hop server)에게 60초마다 자신의 실제주소(NBMA 주소) 등록

```
ip nhrp registration timeout 60
```

```
!
```

//확인

```
R5#ping 10.1.100.2
```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.100.2, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 84/98/116 ms

```
R5#ping 10.1.100.4
```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.100.4, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 92/134/200 ms

• mGRE 터널을 통한 라우팅 설정

//mGRE 터널을 통한 라우팅 설정

<R1>

```
router eigrp 1
network 10.1.10.1 0.0.0.0
network 10.1.12.1 0.0.0.0
!
```

<R2>

```
router eigrp 1
network 10.1.12.2 0.0.0.0
network 10.1.100.2 0.0.0.0
!
```

<R4>

```
router eigrp 1
network 10.1.40.4 0.0.0.0
network 10.1.100.4 0.0.0.0
!
```

<R5>

```
router eigrp 1
network 10.1.50.5 0.0.0.0
network 10.1.100.5 0.0.0.0
!
```

R2#show ip route
(생략)

```
1.0.0.0/24 is subnetted, 1 subnets
C    1.1.23.0 is directly connected,
FastEthernet0/0.23
10.0.0.0/24 is subnetted, 5 subnets
D    10.1.10.0 [90/30720] via 10.1.12.1,
00:00:26, FastEthernet0/0.12
C    10.1.12.0 is directly connected,
FastEthernet0/0.12
D    10.1.40.0 [90/297246976] via
10.1.100.4, 00:00:22, Tunnel0
D    10.1.50.0 [90/297246976] via
10.1.100.5, 00:00:09, Tunnel0
C    10.1.100.0 is directly connected,
Tunnel0
S*   0.0.0.0/0 [1/0] via 1.1.23.3
```

• mGRE 터널을 통한 라우팅 설정

//Split horizon 때문에 R4에는 R5의 내부망 정보가 없음

R4#show ip route

(생략)

1.0.0.0/24 is subnetted, 1 subnets

C 1.1.34.0 is directly connected, FastEthernet0/0.34

10.0.0.0/24 is subnetted, 4 subnets

D 10.1.10.0 [90/297249536] via 10.1.100.2, 00:01:51, Tunnel0

D 10.1.12.0 [90/297246976] via 10.1.100.2, 00:01:51, Tunnel0

C 10.1.40.0 is directly connected, FastEthernet0/0.40

C 10.1.100.0 is directly connected, Tunnel0

S* 0.0.0.0/0 [1/0] via 1.1.34.3

//해결 <R2>

interface tunnel 0

no ip split-horizon eigrp 1

R4#show ip route

(생략)

1.0.0.0/24 is subnetted, 1 subnets

C 1.1.34.0 is directly connected, FastEthernet0/0.34

10.0.0.0/24 is subnetted, 5 subnets

D 10.1.10.0 [90/297249536] via 10.1.100.2, 00:07:37, Tunnel0

D 10.1.12.0 [90/297246976] via 10.1.100.2, 00:07:37, Tunnel0

C 10.1.40.0 is directly connected, FastEthernet0/0.40

D 10.1.50.0 [90/310046976] via 10.1.100.2, 00:00:07, Tunnel0

C 10.1.100.0 is directly connected, Tunnel0

S* 0.0.0.0/0 [1/0] via 1.1.34.3

• mGRE 터널을 통한 라우팅 설정

//확인 명령어

//터널의 목적지 주소와 연결되는 넥스트 홉 IP 주소 (NBMA address)가 저장되어 있음

```
R2#show ip nhrp
```

```
10.1.100.4/32 via 10.1.100.4, Tunnel0 created 00:15:34, expire 01:59:26
```

```
  Type: dynamic, Flags: unique registered
```

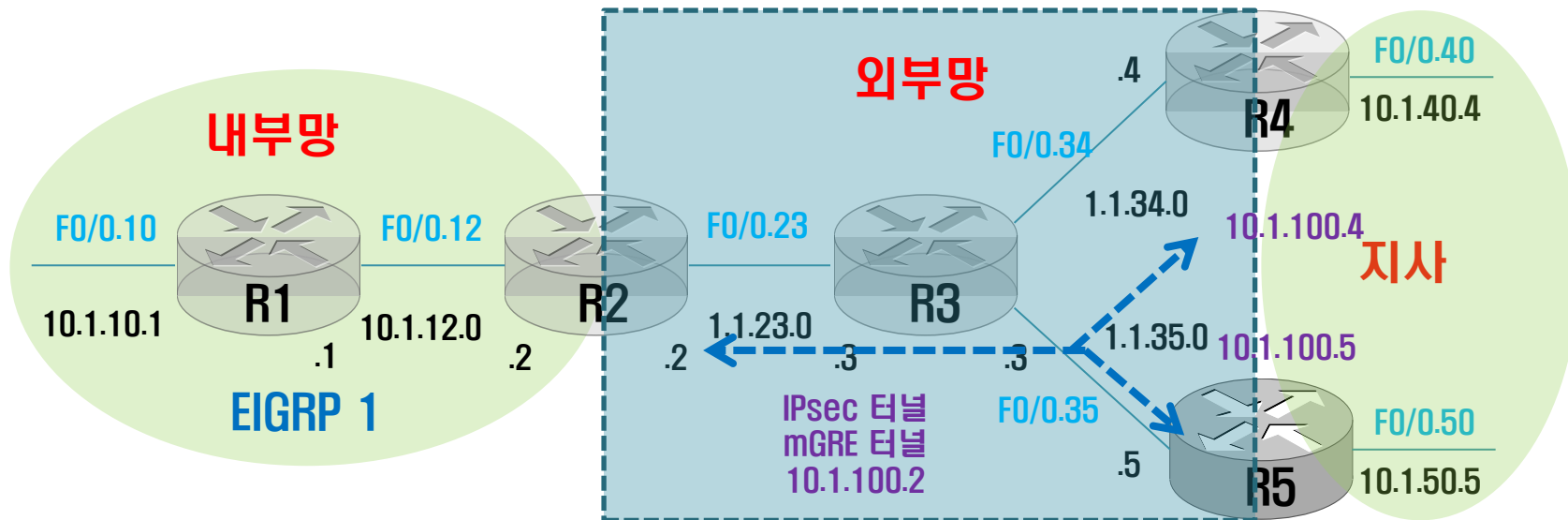
```
  NBMA address: 1.1.34.4
```

```
10.1.100.5/32 via 10.1.100.5, Tunnel0 created 00:15:38, expire 01:59:21
```

```
  Type: dynamic, Flags: unique registered
```

```
  NBMA address: 1.1.35.5
```


• DMVPN 설정



• DMVPN 설정

//DMVPN 설정

<R2>

```
crypto isakmp policy 10
authentication pre-share
!
crypto isakmp key cisco address 0.0.0.0
0.0.0.0
!
crypto ipsec transform-set trset_dmvpn
esp-aes esp-sha-hmac
!
crypto ipsec profile profile_dmvpn
set transform-set trset_dmvpn
!
```

//GRE 터널과 IPsec 프로파일을 연결

```
interface tunnel 0
tunnel protection ipsec profile
profile_dmvpn
!
```

<R4>

```
crypto isakmp policy 10
authentication pre-share
!
crypto isakmp key cisco address 0.0.0.0
0.0.0.0
!
```

```
crypto ipsec transform-set trset_dmvpn
esp-aes esp-sha-hmac
!
crypto ipsec profile profile_dmvpn
set transform-set trset_dmvpn
!
interface tunnel 0
tunnel protection ipsec profile
profile_dmvpn
!
```

<R5>

```
crypto isakmp policy 10
authentication pre-share
!
crypto isakmp key cisco address 0.0.0.0
0.0.0.0
!
crypto ipsec transform-set trset_dmvpn
esp-aes esp-sha-hmac
!
crypto ipsec profile profile_dmvpn
set transform-set trset_dmvpn
!
interface tunnel 0
tunnel protection ipsec profile
profile_dmvpn
!
```

• DMVPN 설정

//확인

R2#show ip route

(생략)

1.0.0.0/24 is subnetted, 1 subnets

C 1.1.23.0 is directly connected, FastEthernet0/0.23

10.0.0.0/24 is subnetted, 5 subnets

D 10.1.10.0 [90/30720] via 10.1.12.1, 00:07:33, FastEthernet0/0.12

C 10.1.12.0 is directly connected, FastEthernet0/0.12

D 10.1.40.0 [90/297246976] via 10.1.100.4, 00:07:27, Tunnel0

D 10.1.50.0 [90/297246976] via 10.1.100.5, 00:00:01, Tunnel0

C 10.1.100.0 is directly connected, Tunnel0

S* 0.0.0.0/0 [1/0] via 1.1.23.3

R2#show crypto isakmp sa

IPv4 Crypto ISAKMP SA

dst	src	state	conn-id	slot	status
1.1.35.5	1.1.23.2	QM_IDLE	1001	0	ACTIVE
1.1.34.4	1.1.23.2	QM_IDLE	1002	0	ACTIVE

IPv6 Crypto ISAKMP SA

• DMVPN 설정

//확인(이어서)

R2#show crypto map

Crypto Map "Tunnel0-head-0" 65536 ipsec-isakmp

Profile name: profile_dmvpn

Security association lifetime: 4608000 kilobytes/3600 seconds

PFS (Y/N): N

Transform sets={
 trset_dmvpn,
}

Crypto Map "Tunnel0-head-0" 65537 ipsec-isakmp

Map is a PROFILE INSTANCE.

Peer = 1.1.34.4

Extended IP access list

access-list permit gre host 1.1.23.2 host 1.1.34.4

Current peer: 1.1.34.4

Security association lifetime: 4608000 kilobytes/3600 seconds

PFS (Y/N): N

Transform sets={
 trset_dmvpn,
}

• EIGRP를 사용한 DMVPN 지사간의 직접 통신

//원격 지사 R4로 가는 경로가 본사 라우터를 통하고 있음

R5#show ip route | i 10.1.40.0

D 10.1.40.0 [90/310046976] via 10.1.100.2, 00:06:02, Tunnel0

//EIGRP 네트워크 광고시 넥스트 홉 IP 주소를 변경하지 않도록 설정

<R2>

```
interface tunnel 0
no ip next-hop-self eigrp 1
!
```

R5#

R5#show ip route | i 10.1.40.0

D 10.1.40.0 [90/310046976] via 10.1.100.4, 00:00:03, Tunnel0

• OSPF를 이용한 DMVPN

<R1 & R2 & R4 & R5>

no router eigrp 1
!

//OSPF 설정

<R1>

```
router ospf 1
network 10.1.10.1 0.0.0.0 area 0
network 10.1.12.1 0.0.0.0 area 0
!
```

<R2>

```
router ospf 1
network 10.1.12.2 0.0.0.0 area 0
network 10.1.100.2 0.0.0.0 area 0
!
```

//지사간 통신을 직접, 장애 발생 시 복구 속도 높이는 Broadcast 사용, DR로 동작하도록

```
interface tunnel 0
ip ospf network broadcast
ip ospf priority 255
!
```

<R4>

```
router ospf 1
network 10.1.40.4 0.0.0.0 area 0
network 10.1.100.4 0.0.0.0 area 0
!
```

//DR이 되지 못하도록 OSPF 우선순위 값 0

```
interface tunnel 0
ip ospf network broadcast
ip ospf priority 0
!
```

<R5>

```
router ospf 1
network 10.1.50.5 0.0.0.0 area 0
network 10.1.100.5 0.0.0.0 area 0
!
```

//DR이 되지 못하도록 OSPF 우선순위 값 0

```
interface tunnel 0
ip ospf network broadcast
ip ospf priority 0
!
```

• OSPF를 이용한 DMVPN

//확인

R2#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.100.4	0	FULL/DROTHER	00:00:38	10.1.100.4	Tunnel0
10.1.100.5	0	FULL/DROTHER	00:00:39	10.1.100.5	Tunnel0
10.1.12.1	1	FULL/BDR	00:00:38	10.1.12.1	FastEthernet0/0.12

R5#show ip route
(생략)

Gateway of last resort is 1.1.35.3 to network 0.0.0.0

1.0.0.0/24 is subnetted, 1 subnets

C 1.1.35.0 is directly connected, FastEthernet0/0.35

10.0.0.0/24 is subnetted, 5 subnets

O 10.1.10.0 [110/11113] via 10.1.100.2, 00:00:11, Tunnel0

O 10.1.12.0 [110/11112] via 10.1.100.2, 00:00:11, Tunnel0

O 10.1.40.0 [110/11112] via 10.1.100.4, 00:00:01, Tunnel0

C 10.1.50.0 is directly connected, FastEthernet0/0.50

C 10.1.100.0 is directly connected, Tunnel0

S* 0.0.0.0/0 [1/0] via 1.1.35.3

• RIPv2를 이용한 DMVPN

<R1 & R2 & R4 & R5>

no router ospf 1
!

//RIPv2 설정

<R1>

```
router rip
version 2
network 10.0.0.0
!
```

<R2>

```
router rip
version 2
network 10.0.0.0
!
```

//RIP 스플릿 호라이즌 비활성화

```
interface tunnel 0
no ip split-horizon
!
```

<R4>

```
router rip
version 2
network 10.0.0.0
!
```

<R5>

```
router rip
version 2
network 10.0.0.0
!
```

• RIPv2를 이용한 DMVPN

//확인

R5#show ip route

*Mar 1 00:31:20.943: %SYS-5-CONFIG_I: Configured from console by console

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is 1.1.35.3 to network 0.0.0.0

1.0.0.0/24 is subnetted, 1 subnets

C 1.1.35.0 is directly connected, FastEthernet0/0.35

10.0.0.0/24 is subnetted, 5 subnets

R 10.1.10.0 [120/2] via 10.1.100.2, 00:00:11, Tunnel0

R 10.1.12.0 [120/1] via 10.1.100.2, 00:00:11, Tunnel0

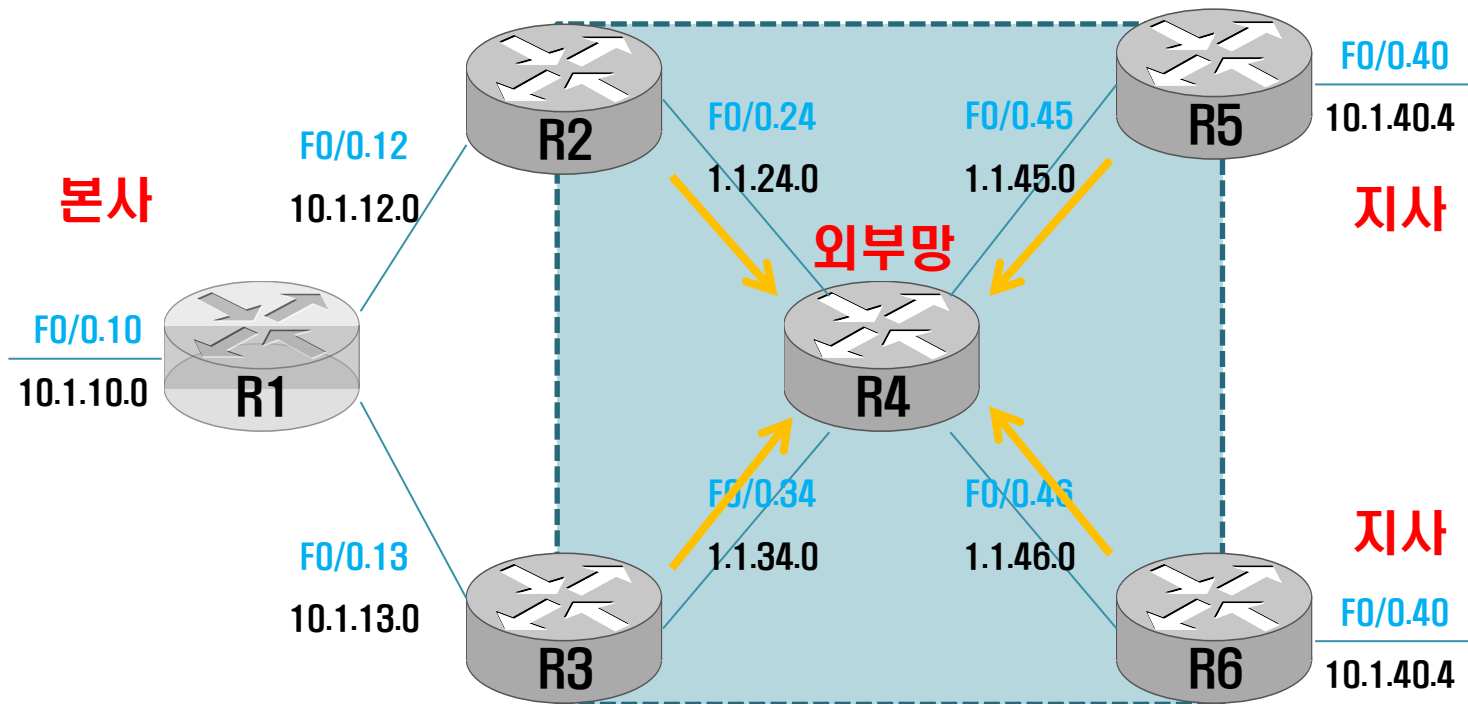
R 10.1.40.0 [120/2] via 10.1.100.4, 00:00:11, Tunnel0

C 10.1.50.0 is directly connected, FastEthernet0/0.50

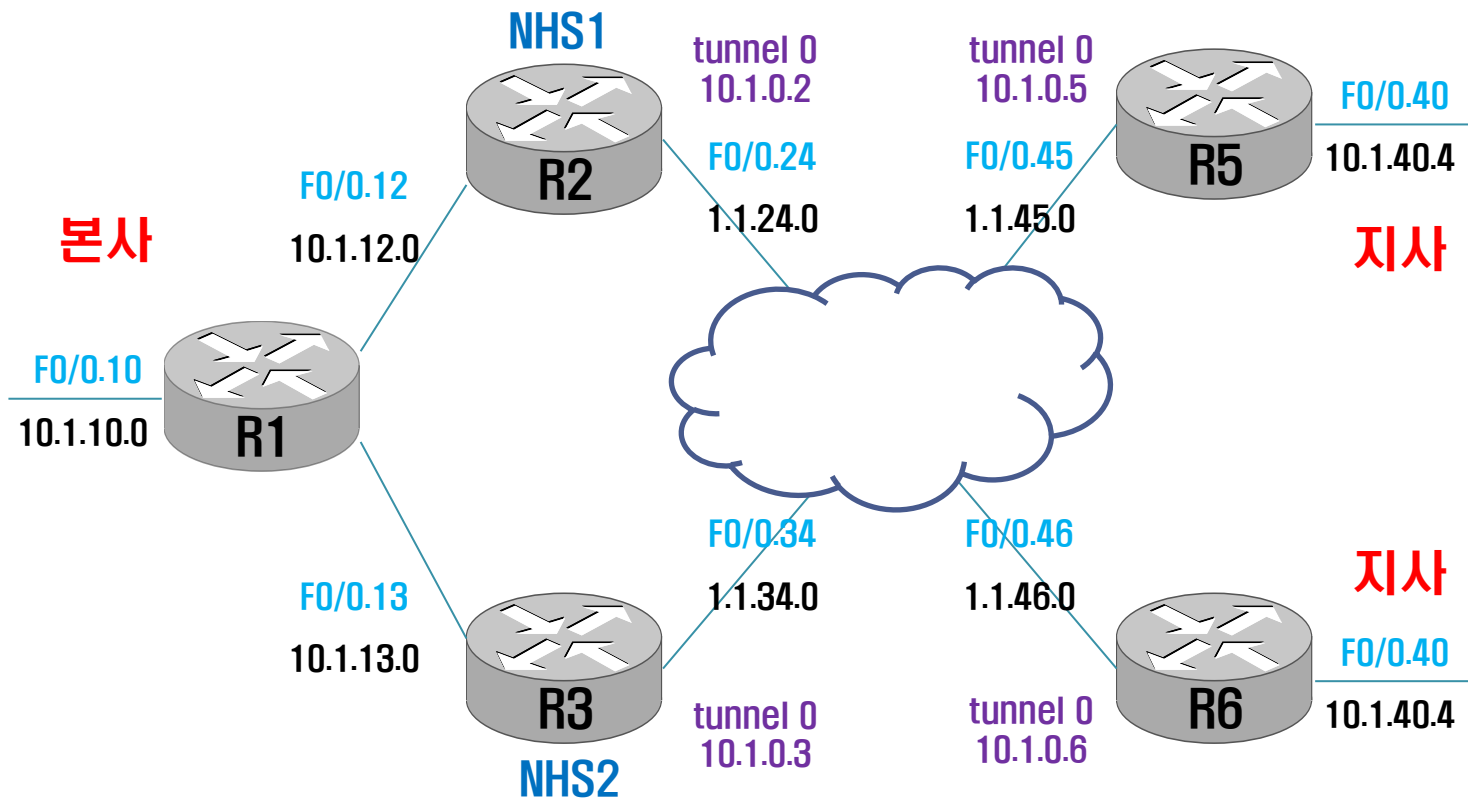
C 10.1.100.0 is directly connected, Tunnel0

S* 0.0.0.0/0 [1/0] via 1.1.35.3

• DMVPN 이중화 테스트 네트워크 구축



• mGRE 터널 구성



• mGRE 터널 구성

//mGRE 터널 구성

//본사 라우터인 R2, R3를 각각의 NHS로
설정

<R2>

```
interface tunnel 0
ip address 10.1.0.2 255.255.255.0
tunnel source 1.1.24.2
tunnel mode gre multipoint
!
ip nhrp network-id 1
!
ip nhrp nhs 10.1.0.3
ip nhrp map 10.1.0.3 1.1.34.3
ip nhrp map multicast 1.1.34.3
!
ip nhrp map multicast dynamic
!
ip nhrp authentication cisco123
ip nhrp holdtime 600
!
```

<R3>

```
interface tunnel 0
ip address 10.1.0.3 255.255.255.0
tunnel source 1.1.34.3
tunnel mode gre multipoint
!
ip nhrp network-id 1
!
ip nhrp nhs 10.1.0.2
ip nhrp map 10.1.0.2 1.1.24.2
ip nhrp map multicast 1.1.24.2
!
ip nhrp map multicast dynamic
!
ip nhrp authentication cisco123
ip nhrp holdtime 600
!
```

• mGRE 터널 구성

//지사 라우터에서는 R2, R3 두 개의 NHS 지정

<R5>

```
interface tunnel 0
ip address 10.1.0.5 255.255.255.0
tunnel source 1.1.45.5
tunnel mode gre multipoint
!
ip nhrp network-id 1
!
ip nhrp nhs 10.1.0.2
ip nhrp map 10.1.0.2 1.1.24.2
ip nhrp map multicast 1.1.24.2
!
ip nhrp nhs 10.1.0.3
ip nhrp map 10.1.0.3 1.1.34.3
ip nhrp map multicast 1.1.34.3
!
ip nhrp authentication cisco123
ip nhrp holdtime 600
ip nhrp registration timeout 120
!
```

<R6>

```
interface tunnel 0
ip address 10.1.0.6 255.255.255.0
tunnel source 1.1.46.6
tunnel mode gre multipoint
!
ip nhrp network-id 1
!
ip nhrp nhs 10.1.0.2
ip nhrp map 10.1.0.2 1.1.24.2
ip nhrp map multicast 1.1.24.2
!
ip nhrp nhs 10.1.0.3
ip nhrp map 10.1.0.3 1.1.34.3
ip nhrp map multicast 1.1.34.3
!
ip nhrp authentication cisco123
ip nhrp holdtime 600
ip nhrp registration timeout 120
!
```


• mGRE 터널 구성

//확인

//ip nhrp registration timeout 120 명령어 사용, 최대 120초 이내에 아래 메시지 표시

R6#debug nhrp packet

NHRP activity debugging is on

//NHS인 R2, R3에게 R6의 터널 종단 주소 10.1.0.6과 연결하려면 패킷을 NBMA 주소, 즉 넥스트 홉 IP주소 1.1.46.6으로 인캡슐레이션해서 전송하면 된다는 것을 등록

*Mar 1 00:44:15.543: NHRP: Send Registration Request via Tunnel0 vrf 0, packet size: 108

*Mar 1 00:44:15.547: src: 10.1.0.6, dst: 10.1.0.2

*Mar 1 00:44:15.551: (F) afn: IPv4(1), type: IP(800), hop: 255, ver: 1

*Mar 1 00:44:15.551: shtl: 4(NSAP), sstl: 0(NSAP)

*Mar 1 00:44:15.555: (M) flags: "unique nat ", reqid: 1

*Mar 1 00:44:15.555: src NBMA: 1.1.46.6

*Mar 1 00:44:15.559: src protocol: 10.1.0.6, dst protocol: 10.1.0.2

*Mar 1 00:44:15.559: (C-1) code: no error(0)

*Mar 1 00:44:15.559: prefix: 255, mtu: 1514, hd_time: 600

*Mar 1 00:44:15.559: addr_len: 0(NSAP), subaddr_len: 0(NSAP), proto_len: 0, pref: 0

*Mar 1 00:44:28.823: NHRP: Send Registration Request via Tunnel0 vrf 0, packet size: 108

*Mar 1 00:44:28.827: src: 10.1.0.6, dst: 10.1.0.3

*Mar 1 00:44:28.827: (F) afn: IPv4(1), type: IP(800), hop: 255, ver: 1

*Mar 1 00:44:28.831: shtl: 4(NSAP), sstl: 0(NSAP)

*Mar 1 00:44:28.831: (M) flags: "unique nat ", reqid: 8

*Mar 1 00:44:28.831: src NBMA: 1.1.46.6

*Mar 1 00:44:28.831: src protocol: 10.1.0.6, dst protocol: 10.1.0.3

*Mar 1 00:44:28.831: (C-1) code: no error(0)

*Mar 1 00:44:28.831: prefix: 255, mtu: 1514, hd_time: 600

*Mar 1 00:44:28.831: addr_len: 0(NSAP), subaddr_len: 0(NSAP), proto_len: 0, pref: 0

• mGRE 터널 구성

//확인

//NHS인 R2, R3가 등록 메시지에 대한 확인 메시지 전송

```
*Mar 1 01:57:28.123: NHRP: Receive Registration Reply via Tunnel0 vrf 0, packet size: 128
*Mar 1 01:57:28.127: (F) afn: IPv4(1), type: IP(800), hop: 255, ver: 1
*Mar 1 01:57:28.127:      shtl: 4(NSAP), sstl: 0(NSAP)
*Mar 1 01:57:28.131: (M) flags: "unique nat ", reqid: 1
*Mar 1 01:57:28.131:      src NBMA: 1.1.46.6
*Mar 1 01:57:28.131:      src protocol: 10.1.0.6, dst protocol: 10.1.0.2
*Mar 1 01:57:28.135: (C-1) code: no error(0)
*Mar 1 01:57:28.139:      prefix: 255, mtu: 1514, hd_time: 600
*Mar 1 01:57:28.139:      addr_len: 0(NSAP), subaddr_len: 0(NSAP), proto_len: 0, pref: 0

*Mar 1 00:44:28.891: NHRP: Receive Registration Reply via Tunnel0 vrf 0, packet size: 128
*Mar 1 00:44:28.895: (F) afn: IPv4(1), type: IP(800), hop: 255, ver: 1
*Mar 1 00:44:28.895:      shtl: 4(NSAP), sstl: 0(NSAP)
*Mar 1 00:44:28.899: (M) flags: "unique nat ", reqid: 8
*Mar 1 00:44:28.899:      src NBMA: 1.1.46.6
*Mar 1 00:44:28.899:      src protocol: 10.1.0.6, dst protocol: 10.1.0.3
*Mar 1 00:44:28.903: (C-1) code: no error(0)
*Mar 1 00:44:28.907:      prefix: 255, mtu: 1514, hd_time: 600
*Mar 1 00:44:28.907:      addr_len: 0(NSAP), subaddr_len: 0(NSAP), proto_len: 0, pref: 0
```

• mGRE 터널 구성

//확인

R2#show ip nhrp

10.1.0.3/32 via 10.1.0.3, Tunnel0 created 00:06:38, never expire

Type: static, Flags: used

NBMA address: 1.1.34.3

10.1.0.5/32 via 10.1.0.5, Tunnel0 created 00:06:38, expire 00:09:25

Type: dynamic, Flags: unique registered

NBMA address: 1.1.45.5

10.1.0.6/32 via 10.1.0.6, Tunnel0 created 00:05:48, expire 00:09:12

Type: dynamic, Flags: unique registered

NBMA address: 1.1.46.6

R3#show ip nhrp

10.1.0.2/32 via 10.1.0.2, Tunnel0 created 01:18:46, never expire

Type: static, Flags: used

NBMA address: 1.1.24.2

10.1.0.5/32 via 10.1.0.5, Tunnel0 created 01:18:42, expire 00:09:04

Type: dynamic, Flags: unique registered

NBMA address: 1.1.45.5

10.1.0.6/32 via 10.1.0.6, Tunnel0 created 01:18:40, expire 00:08:52

Type: dynamic, Flags: unique registered

NBMA address: 1.1.46.6

• mGRE 터널 구성

//확인

R6#ping 10.1.0.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.0.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 28/49/84 ms

R6#ping 10.1.0.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.0.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/86/140 ms

R6#ping 10.1.0.5

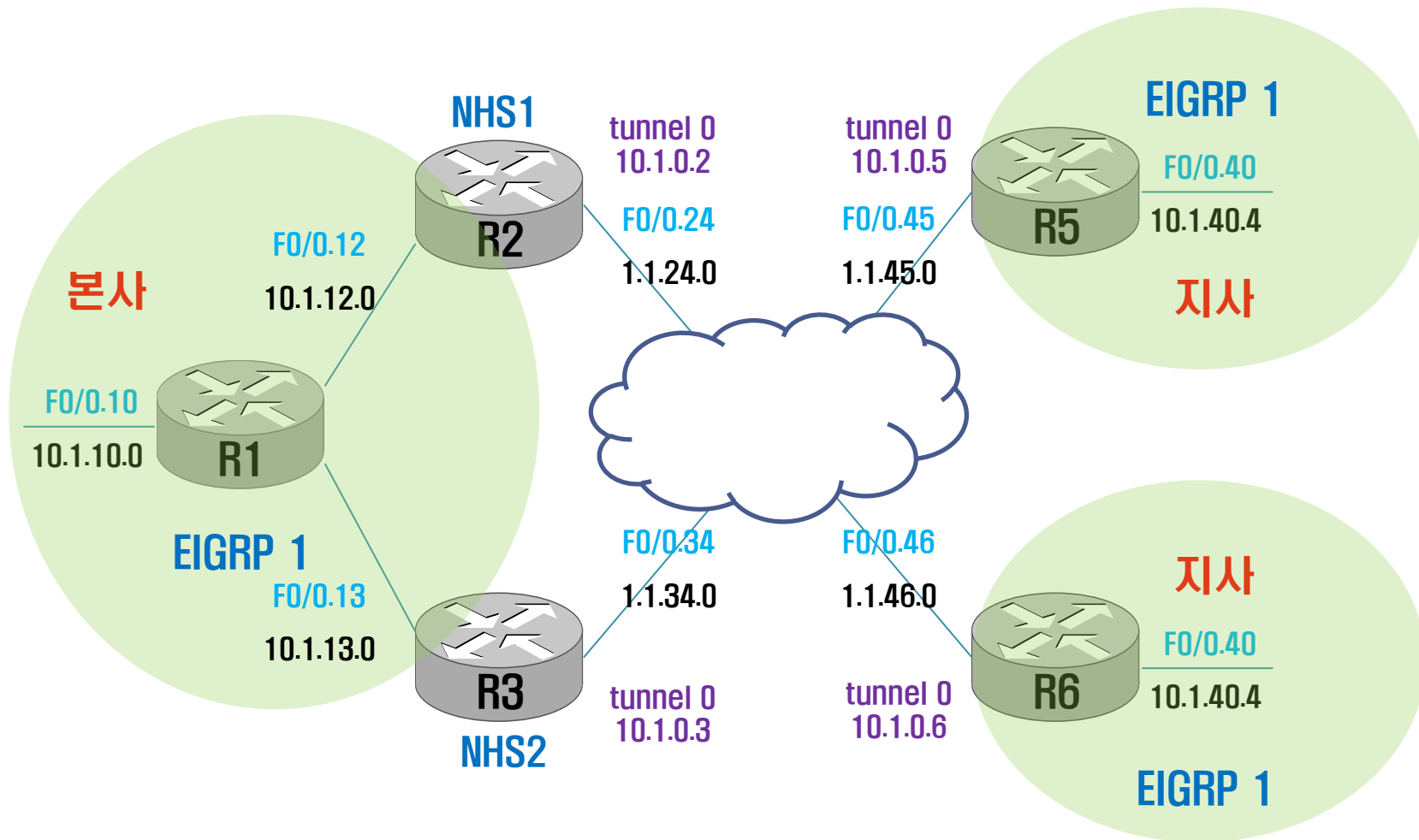
Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.0.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/108/160 ms

- mGRE 터널을 통한 라우팅 설정



• mGRE 터널을 통한 라우팅 설정

//mGRE 터널을 통한 라우팅 설정
//NHS인 R2와 R3에서는 split-horizon
eigrp 1 명령어를 사용하여 EIGRP 스플릿
호라이즌을 비활성화

<R1>

```
router eigrp 1
network 10.1.10.1 0.0.0.0
network 10.1.12.1 0.0.0.0
network 10.1.13.1 0.0.0.0
!
```

<R2>

```
router eigrp 1
network 10.1.12.2 0.0.0.0
network 10.1.0.2 0.0.0.0
!
interface tunnel 0
no ip split-horizon eigrp 1
!
```

<R3>

```
router eigrp 1
network 10.1.13.3 0.0.0.0
network 10.1.0.3 0.0.0.0
!
interface tunnel 0
no ip split-horizon eigrp 1
!
```

<R5>

```
router eigrp 1
network 10.1.50.5 0.0.0.0
network 10.1.0.5 0.0.0.0
!
```

<R6>

```
router eigrp 1
network 10.1.60.6 0.0.0.0
network 10.1.0.6 0.0.0.0
!
```

• mGRE 터널을 통한 라우팅 설정

//확인

```
R1#show ip route eigrp
  10.0.0.0/24 is subnetted, 6 subnets
D    10.1.0.0 [90/297246976] via 10.1.13.3, 00:00:14, FastEthernet0/0.13
      [90/297246976] via 10.1.12.2, 00:00:14, FastEthernet0/0.12
D    10.1.60.0 [90/297249536] via 10.1.13.3, 00:00:14, FastEthernet0/0.13
      [90/297249536] via 10.1.12.2, 00:00:14, FastEthernet0/0.12
D    10.1.50.0 [90/297249536] via 10.1.13.3, 00:00:14, FastEthernet0/0.13
      [90/297249536] via 10.1.12.2, 00:00:14, FastEthernet0/0.12
```

```
R2#show ip route eigrp
  10.0.0.0/24 is subnetted, 6 subnets
D    10.1.10.0 [90/30720] via 10.1.12.1, 00:02:26, FastEthernet0/0.12
D    10.1.13.0 [90/30720] via 10.1.12.1, 00:02:27, FastEthernet0/0.12
D    10.1.60.0 [90/297246976] via 10.1.0.6, 00:02:29, Tunnel0
D    10.1.50.0 [90/297246976] via 10.1.0.5, 00:02:30, Tunnel0
```

//지사 사이의 통신이 모두 본사를 거치고 있음

```
R5#show ip route eigrp
  10.0.0.0/24 is subnetted, 6 subnets
D    10.1.10.0 [90/297249536] via 10.1.0.3, 00:02:49, Tunnel0
      [90/297249536] via 10.1.0.2, 00:02:49, Tunnel0
D    10.1.13.0 [90/297246976] via 10.1.0.3, 00:02:50, Tunnel0
D    10.1.12.0 [90/297246976] via 10.1.0.2, 00:02:49, Tunnel0
D    10.1.60.0 [90/310046976] via 10.1.0.3, 00:02:51, Tunnel0
      [90/310046976] via 10.1.0.2, 00:02:51, Tunnel0
```


• mGRE 터널을 통한 라우팅 설정

//지사 사이의 통신을 직접 이루어지게 하려면,

<R2 & R3>

```
interface tunnel 0
no ip next-hop-self eigrp 1
!
```

R5#show ip route eigrp

10.0.0.0/24 is subnetted, 6 subnets

```
D    10.1.10.0 [90/297249536] via 10.1.0.3, 00:00:11, Tunnel0
      [90/297249536] via 10.1.0.2, 00:00:11, Tunnel0
D    10.1.13.0 [90/297246976] via 10.1.0.3, 00:00:11, Tunnel0
D    10.1.12.0 [90/297246976] via 10.1.0.2, 00:00:11, Tunnel0
D    10.1.60.0 [90/310046976] via 10.1.0.6, 00:00:10, Tunnel0
      [90/310046976] via 10.1.0.6, 00:00:10, Tunnel0
```


•DMVPN설정

//DMVPN 설정

<R2 & R3 & R5 & R6>

```
crypto isakmp policy 1
encryption aes
authentication pre-share
group 2
!
crypto isakmp key cisco address 0.0.0.0 0.0.0.0
!
crypto isakmp keepalive 10
!
crypto ipsec transform-set trset_dmvpn esp-aes esp-md5-hmac
!
crypto ipsec profile profile_dmvpn
set transform-set trset_dmvpn
!
interface tunnel 0
tunnel protection ipsec profile profile_dmvpn
!
```

•DMVPN설정

//확인

```
R5#show ip route eigrp
  10.0.0.0/24 is subnetted, 6 subnets
D       10.1.10.0 [90/297249536] via 10.1.0.3, 00:05:34, Tunnel0
        [90/297249536] via 10.1.0.2, 00:05:34, Tunnel0
D       10.1.13.0 [90/297246976] via 10.1.0.3, 00:05:34, Tunnel0
D       10.1.12.0 [90/297246976] via 10.1.0.2, 00:05:34, Tunnel0
D       10.1.60.0 [90/310046976] via 10.1.0.6, 00:02:05, Tunnel0
        [90/310046976] via 10.1.0.6, 00:02:05, Tunnel0
```

//R2의 tunnel 0를 shutdown

```
*Mar  1 00:07:10.995: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor 10.1.0.2 (Tunnel0)
is down: holding time expired
```

//모든 경로가 R3로 변경

```
R5#show ip route eigrp
  10.0.0.0/24 is subnetted, 6 subnets
D       10.1.10.0 [90/297249536] via 10.1.0.3, 00:00:06, Tunnel0
D       10.1.13.0 [90/297246976] via 10.1.0.3, 00:05:48, Tunnel0
D       10.1.12.0 [90/297249536] via 10.1.0.3, 00:00:06, Tunnel0
D       10.1.60.0 [90/310046976] via 10.1.0.6, 00:00:06, Tunnel0
```

•DMVPN설정

//확인

//R2의 tunnel 0를 no shutdown

*Mar 1 00:09:08.627: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor 10.1.0.2 (Tunnel0) is up: new adjacency

//경로가 R2, R3로 부하분산

R5#show ip route eigrp

10.0.0.0/24 is subnetted, 6 subnets

```
D    10.1.10.0 [90/297249536] via 10.1.0.3, 00:00:01, Tunnel0
      [90/297249536] via 10.1.0.2, 00:00:01, Tunnel0
D    10.1.13.0 [90/297246976] via 10.1.0.3, 00:00:01, Tunnel0
D    10.1.12.0 [90/297246976] via 10.1.0.2, 00:00:01, Tunnel0
D    10.1.60.0 [90/310046976] via 10.1.0.6, 00:00:01, Tunnel0
      [90/310046976] via 10.1.0.6, 00:00:01, Tunnel0
```